

Robot finger that is more sensitive than human's

Robots can be given the sense of feel. All they need is the right sensors, actuators and software to be able to at least identify different materials by touch—according to researchers at the University of Southern California's Viterbi School of Engineering.



Robot with a tactile sensor built to mimic the human fingertip

In a study published in *Frontiers in Neurobotics*, researchers showed that a specialised robot can outperform humans in identifying different materials by touch. The research paves the way for advancements in prostheses, personal assistive robots and consumer product testing.

"The robot was equipped with

a new type of tactile sensor built to mimic the human fingertip. It also used a newly designed algorithm to make decisions about how to explore the outside world by imitating human strategies. Capable of other human sensations, the sensor can also tell where and in which direction forces are applied to the fingertip and even the thermal properties of an object being touched," the study said.

Like the human finger, the BioTac sensor developed by researchers has a soft, flexible skin over a liquid filling. The skin has fingerprints on its surface which help in enhancing its sensitivity to vibration. As the finger slides over a textured surface, the skin vibrates in characteristic ways. These vibrations are then detected whenever the fingertip touches the surface. Though the human finger uses similar vibrations to identify textures, the robot finger is said to be even more sensitive with almost 95 per cent efficiency.

Glowing light bulbs floating mid-air

A thousand glowing light bulbs floating in mid-air could now be a possibility, thanks to a project by Chris Rieger, an electrical engineering student at University of Queensland.

Rieger used a combination of wireless power transfer and magnetic levitation technologies. While he modified an existing circuit built by Eric Taylor for levitating the bulb, he made changes to the circuit design by Marko for building a wireless power transfer.

Rieger said, "The high-frequency magnetic field generated from the wireless power transfer circuit does not affect the levitation aspect of the bulb." For the levitation system, he opted for a 1.325mV/g sensor, as the magnetic flux it was sensing was maxing out the reading on others.

World's thinnest screen made using soap bubble

Researchers at the University of Tokyo have developed the world's thinnest screen, using a soap bubble! This screen is different from regular screens because a soap bubble is transparent and the micro-membrane allows the passage of light and displays colours.

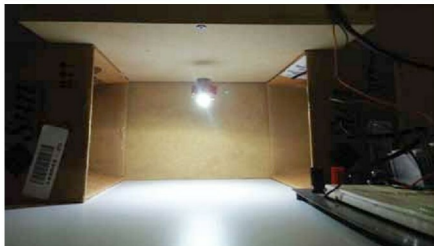
Researcher Yoichi Ochiai along with his colleagues created an ultrathin and flexible bidirectional reflectance distribution function. They used a mix of two colloidal liquids to make the screen. Though a mixture of liquids is used in the display, soap is the main ingredient.

Ochiai's official blog states that the researchers can control the surface, texture and reflectance of the surface by using speakers that emit ultrasonic sound waves. The combination of the ultrasonic waves and ultra-thin membranes results in more realistic, distinctive and vivid imageries on the screen. Interestingly, one can even poke the bubble allowing for interactivity.

If the researchers' claims are anything to go by, this screen will open a new chapter for flexible displays.

Gigapixel camera

What is the highest resolution that avid photographers have ever dreamt of?



Levitating light bulb